

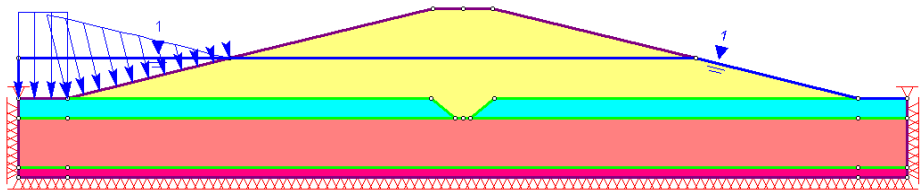
# SSR Search Area

In this tutorial, *Phase2* is used to determine the factor of safety of an embankment using the shear strength reduction (SSR) method. The slope stability analysis is restricted to one side of the embankment by defining an SSR search area.

## Topics covered

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- Import from *Slide*
- Shear strength reduction
- SSR search area
- Add failure surface from *Slide*



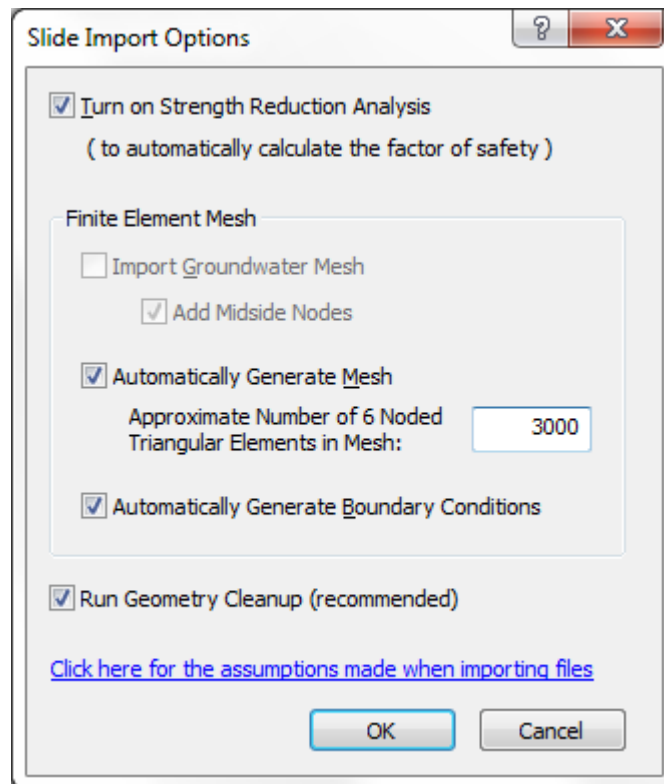
## Model

In this tutorial we will import a model that was constructed in *Slide*. *Slide* is a 2D limit equilibrium slope stability program available from Rocscience. If you are not familiar with *Slide*, see the Rocscience website for information.

To import the model file into *Phase2*, select Import from the File menu and click on Import Slide. Open the file **Tutorial 10 Slide File.slim** found in the Examples > Tutorials folder in your *Phase2* installation directory.

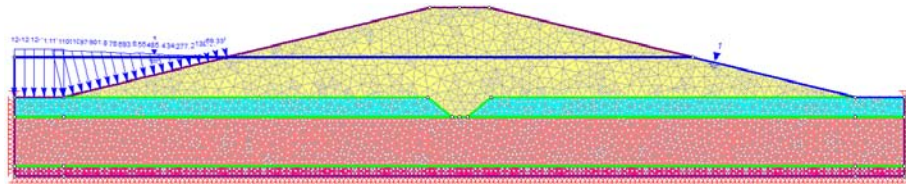
TIP: You can also import a *Slide* file by simply choosing Open from the File menu. At the bottom of the Open dialog, for Files of type, select Slide File Format (\*.slim) from the drop down menu.

Once you have opened the *Slide* file, you will see the following dialog.



This allows you to set various options for the finite element analysis. We want to perform a shear strength reduction (SSR) analysis to determine the factor of safety for slope stability so leave this option on. We also want *Phase2* to automatically generate a finite element mesh and appropriate boundary conditions so leave these options on as well (*Slide* analyses do not require a finite element mesh so the mesh must be generated by *Phase2*). Click OK to accept the defaults.

You should now see a model that looks like this.



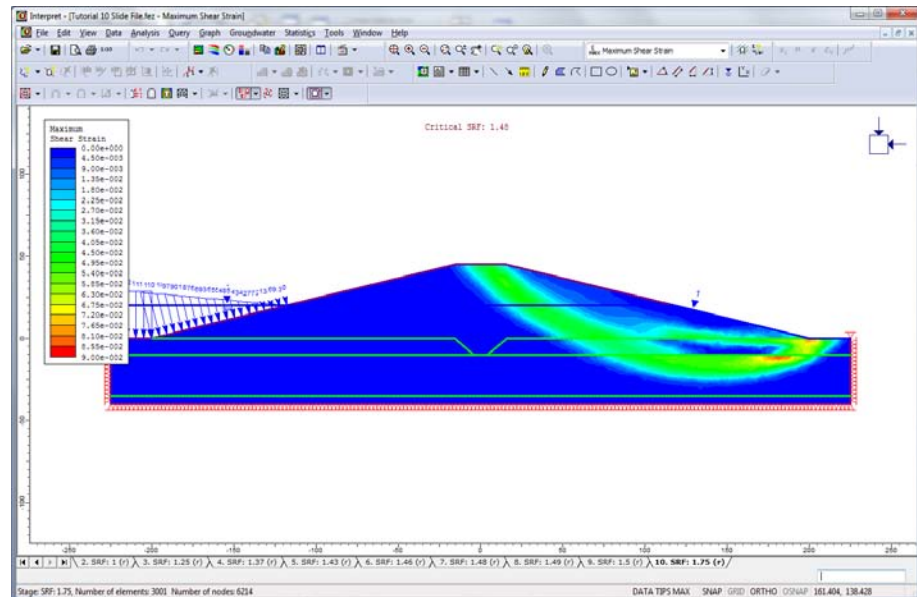
This model represents a sloped embankment built upon a sand layer overlying a foundation clay layer that sits on top of bedrock. The water table is 25 feet below the top of the embankment and ponded water exists at the left side of the model. The ponded water exerts a force on the soil as shown by the blue arrows.

## SSR Analysis – Default Region

The model is already completely built so we can perform the shear strength reduction analysis. Save the file in *Phase2* format by selecting Save As from the File menu. Now select Compute from the Analysis menu. The computation will take a few minutes. When it is finished click on the Interpret button to see the results.



You will now see a plot showing the maximum shear strain in the model for the critical shear strength reduction factor (SRF = 1.48). If you click on the tabs for higher SRF factors, you will see the development of a clear failure surface on the right side of the embankment. The plot for SRF = 1.75 is shown below.



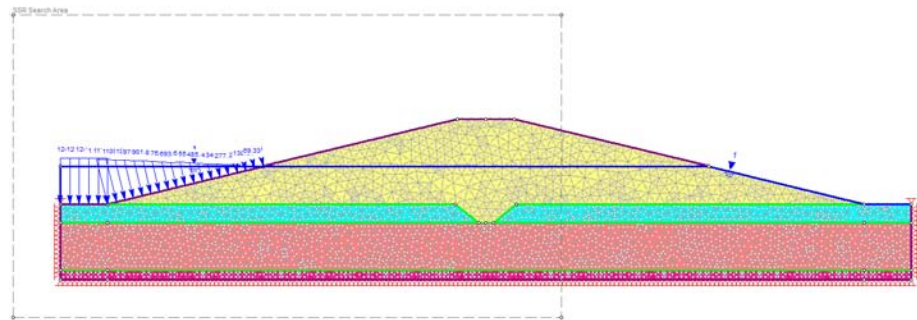
If you display the deformation vectors  you will clearly see the displacement of the slope on the right side with very little deformation on the left side.

In an SSR analysis, *Phase2* automatically finds the slope failure mechanisms associated with the minimum shear strength reduction (minimum factor of safety). In this case, the most likely failure occurs on the right side of the embankment. What if we are only interested in failure on the left side? We need to somehow restrict the analysis to the left side of the model. In *Slide*, you can simply choose the failure to proceed from right to left. In *Phase2* we can define an SSR search area.

## Define SSR Search Area



Go back to the *Phase2* model program by clicking on the Open Modeller button. We will now specify the area over which we would like to perform the SSR analysis. Go to the **Analysis** menu and select **SSR Search Area > Define SSR Search Area(window)**. You will now see cross hairs and you will be asked to enter the first point of the search area. You will need to enter two points to define two diagonally opposite corners of a rectangular area. We wish to encompass the left side of the model so choose a first point above and to the left of the model (near -250, 100). The exact coordinates do not matter so you can use the mouse to choose the point). Now choose a second point below the model and just to the right of the embankment plateau (around 40, -60). Be sure to include the entire flat top of the embankment since it is likely that the failure surface will intersect somewhere on this plateau. The model should now look like this.

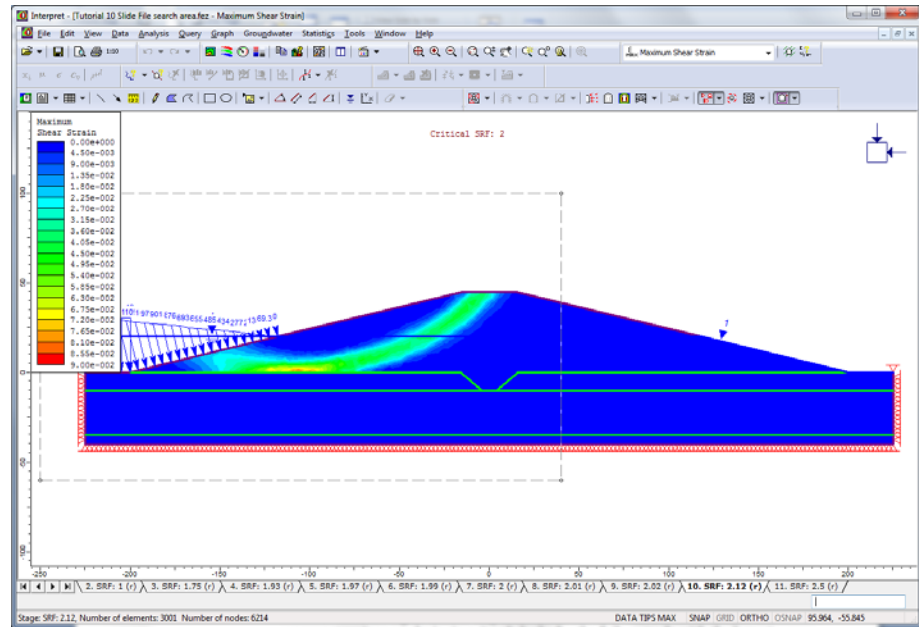


You can easily resize the search area by right clicking on one of the corner points and clicking on Move To. You will now see the cross hairs again and you can move the point to a new location. Click the left mouse button to finish the move. You can delete the search area by going to the Analysis menu and selecting **SSR Search Area > Delete SSR Search Area**.

You have now finished defining the SSR search area. Save the file with a different name and run Compute. Open the Interpreter.

## Interpret

A screen is now displayed with the critical strength reduction factor (SRF) of 2.0 written at the top of the window. Click through the higher SRF plots and you will see the development of a failure zone on the left side. The plot below shows the maximum shear strain for SRF = 2.12.

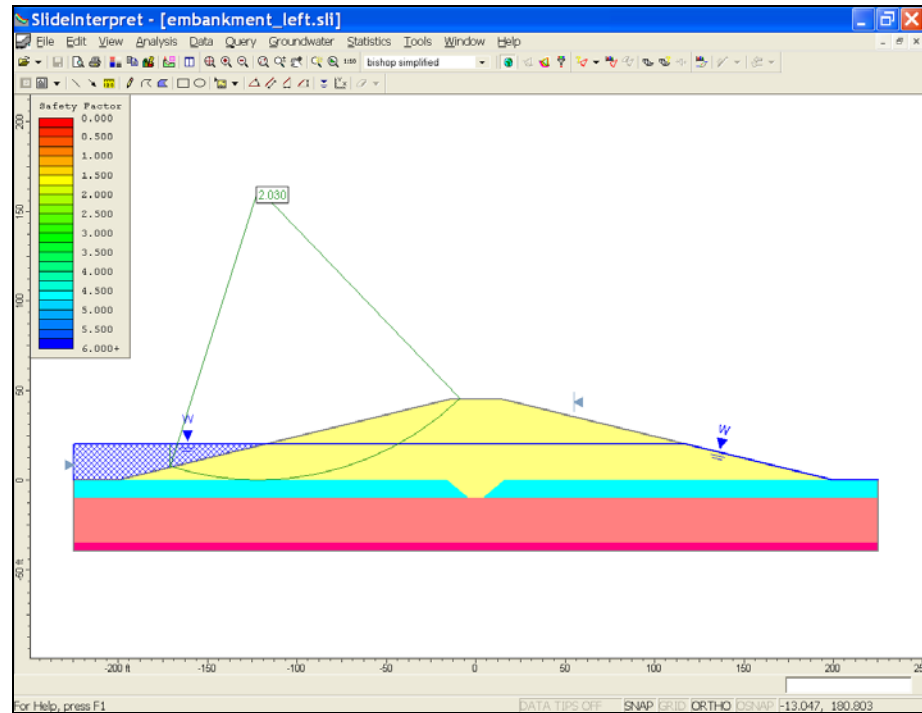


Note that the critical SRF for this analysis (2.0) is higher than the critical SRF of the previous analysis with no search area defined (SRF = 1.48). This indicates that the left side of the embankment is more stable than the right. This is why *Phase2* did not show failure on the left in the previous analysis since it only looks for the failure mode with the lowest factor of safety, unless you restrict the search area as we have done here.

The reason for the different stability of the right and left sides, is due to the stabilizing effect of the ponded water force on the left side of the embankment.

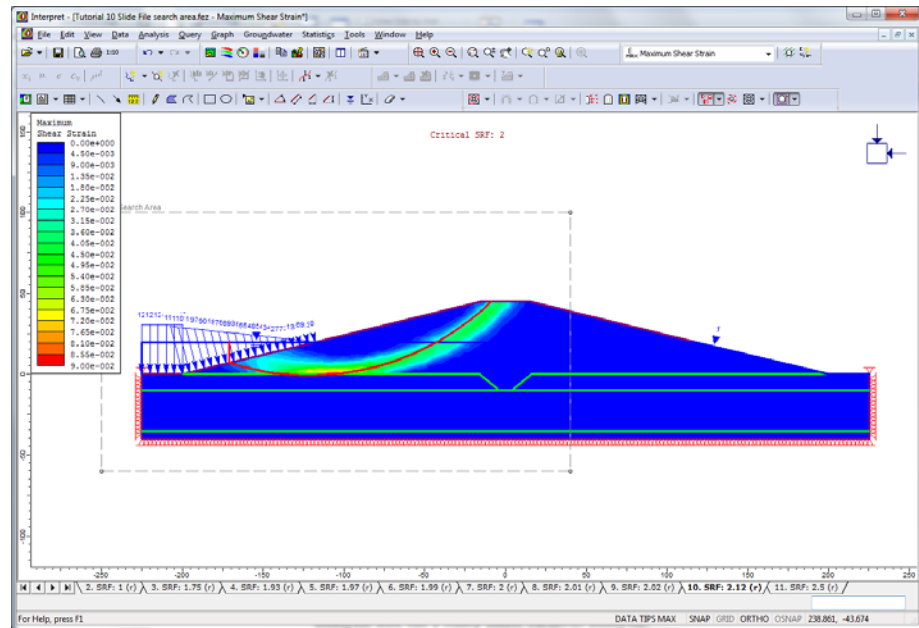
**TIP:** you can animate the viewing of the different SRF stages by choosing Animate Tabs from the Data menu. Hit the Esc key to stop the animation.

If you are a *Slide* user, you can directly compare the *Slide* analysis with the *Phase2* analysis. If you run the *Slide* model and you are looking at the Interpret window you will see a circular failure surface as shown.



Note the factor of safety calculated by *Slide* is 2.03, which is very close to the value calculated by *Phase2* (2.0).

In the Slide Interpreter, right click on the failure surface and select **Copy (Slide modeler format)**. Go back to the *Phase2* Interpret window and choose the plot for SSR: 2.12. Go to the **Edit** menu and choose **Paste from Slide Interpreter**. You will now see the failure surface appear on the plot as a black line. You can modify the appearance of the line by right clicking on it and choosing **Format**. Change the colour to red and the weight to 3. Click OK and your plot should now look like this.



You can see the very good agreement between the *Slide* limit equilibrium analysis and the *Phase2* finite element analysis.

This concludes the tutorial, you may now exit the *Phase2* Interpret and *Phase2* Model programs.